

使用 ADK 和 Google 地圖基礎建構行程規劃代理

來源：<https://codelabs.developers.google.com/next26/maps-grounding?hl=zh-tw>

步驟數：10

建構以 Google 地圖建立基準為基礎的日行程規劃代理程式。

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步驟 1 · 約 0 分鐘

1. 簡介

在本程式碼研究室中，您將使用 Agent Development Kit (ADK) 建構旅遊規劃代理，並以 Google 地圖為基礎。您將**提示**代理程式生成風景優美的路線和餐廳建議，並運用 Google 地圖的真實世界資料。

學習內容

- 使用 **Agent Starter Pack** 初始化代理程式專案
- 設定代理程式以使用 **Google 地圖建立基準** 工具
- 使用網頁介面在本機測試產生的代理程式

軟硬體需求

- 網路瀏覽器，例如 [Chrome](#)
- 已啟用計費功能的 Google Cloud 雲端專案

本程式碼研究室適合中階開發人員，也就是對 Python 和 Google Cloud 有些瞭解，但不一定是專家。

步驟 2 · 約 5 分鐘

2. 事前準備

info

從做中學 免付費

立即取得 Google Cloud 抵免額，並完成這個實驗室。無須提供信用卡或付款方式。

啟用

建立 Google Cloud 專案

1. 在 [Google Cloud 控制台](#) 的專案選取器頁面中，[選取或建立 Google Cloud 專案](#)。
2. 確認 Cloud 專案已啟用計費功能。瞭解如何[檢查專案是否已啟用計費功能](#)。

啟動 Cloud Shell

1. 驗證：

```
gcloud auth list
```

2. 確認專案：

```
gcloud config get project
```

3. 視需要設定：

```
export PROJECT_ID=<YOUR_PROJECT_ID>
gcloud config set project $PROJECT_ID
```

啟用 API

執行下列指令，啟用所有必要的 API：

```
gcloud services enable \
  aiplatform.googleapis.com
```

3. 安裝 Agent Starter Pack

如要開始 ADK 專案，最簡單的方法就是使用 Agent Starter Pack。Google Cloud Agent Starter Pack 是一項開放原始碼指令列介面 (CLI) 工具，旨在加速在 Google Cloud 上開發及部署可供正式環境使用的生成式 AI 代理程式。

1. 確認已安裝 `uv`，然後執行建立指令來初始化新的代理程式專案：

```
uvx agent-starter-pack create
```

如果沒有安裝 `uv`，可以使用 `pip`：

```
# Create and activate a Python virtual environment
python -m venv .venv && source .venv/bin/activate

# Install the agent starter pack
pip install --upgrade agent-starter-pack

# Create a new agent project
agent-starter-pack create
```

詳情請參閱 [agent-starter-pack 存放區](#)。

2. 系統出現提示時，請提供下列選項，為使用 React 前端的本機開發作業設定專案：

- 代理程式範本：`adk` (簡單的 React 代理程式)
- 部署：`none` (目前已停用 Cloud Deploy)
- **Region** (區域)：`us-central1`

這會產生專案目錄結構，其中包含主要代理程式邏輯、測試和 `GEMINI.md` 指南。前往新目錄：

```
cd my-agent
```

步驟 4 · 約 2 分鐘

4. 設定依據

Agent Starter Pack 會生成 `GEMINI.md` 檔案，指示 AI 輔助程式碼編寫工具如何管理專案。我們將更新這項內容，加入 Google 地圖 Grounding 說明文件。

1. 在編輯器中開啟 `GEMINI.md`。
2. 在 `## Reference Documentation` 區段下方新增下列參照連結：

```
- **Google Maps Grounding**: https://docs.cloud.google.com/vertex-ai/generative-ai/docs/grounding/grounding-with-google-maps
```

這項脈絡有助於 AI 程式設計助理瞭解建立基準功能。

5. 更新代理程式

現在我們要設定代理程式，讓它擔任行程規劃師，並使用 Google 地圖 Grounding 工具。

1. 開啟 `app/agent.py` 檔案。
2. 將 `app/agent.py` 的所有內容替換為下列程式碼：

```
"""Agent application for the itinerary planner codelab."""

import os
import google.auth
from google.adk.agents import Agent
from google.adk.apps import App
from google.adk.models import Gemini
from google.adk.tools import google_maps_grounding
from google.genai import types

# Authenticate and set environment variables
_, project_id = google.auth.default()
os.environ["GOOGLE_CLOUD_PROJECT"] = project_id
os.environ["GOOGLE_CLOUD_LOCATION"] = "global"
os.environ["GOOGLE_GENAI_USE_VERTEXAI"] = "True"

# Define the root agent
root_agent = Agent(
    name="itinerary_planner_agent",
    model=Gemini(
        model="gemini-2.5-flash",
        retry_options=types.HttpRetryOptions(attempts=3),
    ),
    instruction=(
        "You are an itinerary planner agent. Help users plan their trips by"
        " recommending restaurants and scenic routes. Use the"
        " google_maps_grounding tool to get both restaurant recommendations and"
        " route recommendations based on user preferences. When calling for"
        " restaurant recommendation, prompt the tool to tell you about the vibe"
        " of the place. When calling for routes with multiple legs, describe"
        " each of those legs with a brief sentence. Always describe the key"
        " landmarks along the route in one brief sentence."
    ),
    # Add the Google Maps Grounding tool to the agent
    tools=[google_maps_grounding],
)

app = App(
    root_agent=root_agent,
    name="app",
)
```

這段程式碼會設定 `gemini-2.5-flash` 型代理程式，使用 `google_maps_grounding` 工具擷取地點和路線的最新資訊。

如要查看所有可用模型，請參閱 [Vertex AI 說明文件](#)。

步驟 6 · 約 5 分鐘

6. 執行代理程式

代理程式邏輯設定完成後，請嘗試在本機網頁介面中測試。

1. 在 `my-agent` 目錄的根層級，執行下列指令來啟動網頁應用程式：

```
uv run adk web
```

或使用虛擬環境：

```
adk web
```

2. 在瀏覽器中開啟終端機輸出內容中提供的網址。
3. 提出問題來測試代理程式。例如：

- 「規劃舊金山一日遊行程，包括一間好吃的義大利餐廳。」
- 我要去東京，請提供行程，包含有趣的歷史地標和評價很高的拉麵店，氣氛要溫馨。

您應該會看到類似詳細行程的輸出內容，其中直接從 Google 地圖擷取真實評論和路線說明。

role: "model"

groundingMetadata:

groundingChunks:

0:

maps:

placeId: "places/ChIJ3j8pPPCAYAR5VtEFCtFBw"

text: "++About++ ++Type++ Italian Restaurant + ++Business Status++ Operational + ++Address++ 716 Columbus Ave, San Francisco, CA 94133, USA + ++Location++ 37.801687, -122.411774 + ++Open Now++ Yes + ++Rating++ 4.6 (3265 reviews) + ++Price Level++ Inexpensive + ++Phone++ (415) 712-8874 + ++Summary++ Italian deli & market stocking housemade pasta, imported cheese, charcuterie & more in honey dips. + ++Additional Summary++ Informal restaurant turning out homemade pasta dishes such as gnocchi and lasagna, plus classic Italian mains and pizzas. + ++Offers Takeout++ Yes + ++Offers Delivery++ Yes + ++Offers Dine-in++ Yes + ++Good for Children++ Yes + ++Has Restroom++ Yes + ++Outdoor Seating++ Yes + ++Live Music++ No + ++Menu for Children++ No + ++Serves Cocktails++ No + ++Serves Desserts++ Yes + ++Serves Coffee++ Yes + ++Has Wifi++ No + ++Good for Matching Sports++ No + ++Serves Lunch++ Yes + ++Serves Dinner++ Yes + ++Serves Beer++ Yes + ++Serves Wine++ Yes + ++Serves Vegetarian Food++ Yes ++Opening Hours (Local time):++ ++ Monday: 11:00 AM - 9:00 PM ++ Tuesday: 11:00 AM - 9:00 PM ++ Wednesday: 11:00 AM - 9:00 PM ++ Thursday: 11:00 AM - 9:00 PM ++ Friday: 11:00 AM - 10:00 PM ++ Saturday: 11:00 AM - 10:00 PM ++ Sunday: 11:00 AM - 9:00 PM ++Nearby Landmarks & Areas:++ ++ Down the road from Washington Square ++ Around the corner from Joe DiMaggio Playground Park ++ Down the road from XDX Truffles ++ Across the road from Graffeo Coffee Roasting Co. ++ Around the corner from North Beach Farmers Market ++ On the outskirts of North Beach ++ On the outskirts of Parking lot ++ Near Parking lot ++Parking options++ ++ Paid street parking:++ Yes ++Accessibility++ ++ Wheelchair accessible entrance:++ Yes ++ Wheelchair accessible seating:++ Yes ++Payment options++ ++Credit Card:++ Yes ++Debit Card:++ Yes ++Cash Only:++ No ++AFC:++ Yes title: "The Italian Homemade Company" url: "https://maps.google.com/?cid=14085998629817082121"

1:

maps:

placeId: "places/ChIJx2MBuHW_j4ARk1f07KSqy0E"

text: "++About++ ++Type++ Italian Restaurant + ++Business Status++ Operational + ++Address++ 1132 Valencia St, San Francisco, CA 94110, USA + ++Location++ 37.754715, -122.421194 + ++Open Now++ Yes + ++Rating++ 4.4 (1057 reviews) + ++Price Level++ Moderate + ++Phone++ (415) 605-9848 + ++Summary++ Classic Italian fare including pizza, pasta, and meat dishes are offered at this modest restaurant. + ++Additional Summary++ Diner-like Italian eatery with a salad bar and pizza on the menu, plus wine and cocktails. + ++Takes Reservations:++ Yes + ++Offers Takeout:++ Yes + ++Offers Delivery:++ Yes + ++Offers Dine-in:++ Yes + ++Good for Children:++ Yes + ++Has Restroom:++ Yes + ++Good for Groups:++ Yes + ++Outdoor Seating:++ Yes + ++Live Music:++ No + ++Menu for Children:++ Yes + ++Serves Cocktails:++ Yes + ++Serves Desserts:++ Yes + ++Serves Coffee:++ Yes + ++Has Wifi:++ Yes + ++Good for Matching Sports:++ No + ++Serves Lunch:++ Yes + ++Serves Dinner:++ Yes + ++Serves Beer:++ Yes + ++Serves Wine:++ Yes + ++Serves Brunch:++ Yes + ++Serves Vegetarian Food:++ Yes ++Opening Hours (Local time):++ ++ Monday: 11:00 AM - 11:00 PM ++ Tuesday: 11:00 AM - 11:00 PM ++ Wednesday: 11:00 AM - 11:00 PM ++ Thursday: 11:00 AM - 11:00 PM ++ Friday: 11:00 AM - 12:00 AM ++ Saturday: 11:00 AM - 12:00 AM ++ Sunday: 11:00 AM - 11:00 PM ++Nearby Landmarks & Areas:++ ++ Across the road from City College of San Francisco - Mission Center ++ Down the road from The Salvation Army Mission Corps Community Center ++ Beside Deccan House ++ Beside U Save Plumbing & Hardware ++ Across the road from El Valenciano ++ On the outskirts of Mission District ++ On the outskirts of Parking lot ++ On the outskirts of Mission Campus

SESSION ID 2fb64127-c8f3-4e39-0e0b-123fd672dd0f USER ID user

Token Streaming | + New Session

#1

Plan a 1-day itinerary in San Francisco including a good Italian restaurant.

Here is a possible 1-day itinerary for San Francisco:

Morning:

Start your day with a visit to the iconic **Golden Gate Bridge**. Enjoy the breathtaking views of the Pacific Ocean, the city skyline, and Alcatraz Island. Afterwards, head to **Fisherman's Wharf** to experience its lively atmosphere, watch the sea lions at Pier 39, and perhaps grab a clam chowder in a sourdough bowl.

Lunch:

For lunch, I recommend **The Italian Homemade Company** in North Beach. This informal restaurant is known for its homemade pasta dishes, including gnocchi and lasagna, as well as classic Italian mains and pizzas. It has a great rating of 4.6 stars based on over 3,200 reviews.

Afternoon:

After your delicious lunch, explore the charming **North Beach** neighborhood. Wander through its streets, discover local shops, and then make your way to **Coit Tower** for panoramic views of the city. Finally, take a stroll through the vibrant streets of **Chinatown**, the oldest Chinatown in North America, and immerse yourself in its unique culture and history.

To help you navigate, here are some directions for your trip: Here is a possible 1-day itinerary for San Francisco, including a recommendation for a good Italian restaurant:

Morning: Iconic Views & Waterfront Charm Start your day at the iconic **Golden Gate Bridge**. You can reach it by car, public transport, or rideshare from anywhere in San Francisco. Enjoy the breathtaking views of the Pacific Ocean, the city skyline, and Alcatraz Island.

Next, head to **Fisherman's Wharf**, a short drive from the Golden Gate Bridge. Here, you can experience the lively atmosphere, watch the famous sea lions at Pier 39, and perhaps grab a clam chowder in a sourdough bowl.

Lunch: Authentic Italian in North Beach For lunch, I recommend **The Italian Homemade Company** (716 Columbus Ave, San Francisco, CA 94133). This informal restaurant is known for its homemade pasta dishes, such as gnocchi and lasagna, as well as classic Italian mains and pizzas. It has a great rating of 4.6 stars based on over 3,200 reviews and a casual, homey vibe.

If you are walking from Fisherman's Wharf to The Italian Homemade Company, the distance is approximately 911 meters, and it will take about 13.3 minutes.

Afternoon: City Panoramas & Cultural Exploration After your delicious Italian lunch, explore the charming **North Beach** neighborhood. From The Italian Homemade Company, it's a pleasant walk to **Coit Tower** (1 Telegraph Hill Blvd, San Francisco, CA 94133). This walk is about 673 meters and should take approximately 13.3 minutes. At Coit Tower, you can enjoy panoramic views of the city, including Alcatraz, the Golden Gate Bridge, and the Bay Bridge.

Finally, take a stroll through the vibrant streets of **Chinatown**. From Coit Tower, it's about a 975-meter walk, taking roughly 15.8 minutes. Immerse yourself in the oldest Chinatown in North America, with its unique culture, shops, and architecture.

Click here

#2

Type a Message...

Cannot find specified session. Creating a new one. OK

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7. 驗證程式碼中的基礎

如要以程式輔助方式確認代理程式是否已成功使用 Maps 建立基準，可以檢查回應事件的 Maps 專屬中繼資料。

執行代理程式 (例如在測試腳本中) 時，代理程式會產生包含 `grounding_metadata` 的事件。您可以疊代這個中繼資料中的 `grounding_chunks`，並檢查 `maps` 屬性。

以下範例說明如何檢查 `maps` 屬性，與您在自動化測試中使用的類似：

```
async for event in runner.run_async(
    user_id="test_user",
    session_id=session.id,
    new_message=content,
):
    if event.grounding_metadata:
        if event.grounding_metadata.grounding_chunks:
            for chunk in event.grounding_metadata.grounding_chunks:
                # Check for the maps attribute to confirm maps grounding
                if hasattr(chunk, "maps") and chunk.maps:
                    print("SUCCESS: Maps grounding chunks detected in the response!")
```

8. 擷取編碼折線

除了驗證是否已建立基準，您可能還想擷取特定資料，例如路線路徑。地圖基礎工具傳回路線資訊時，通常會包含「編碼折線」，可用於在地圖前端算繪路線。

如要找出這條折線，請檢查 `grounding_chunks` 的 `maps` 屬性中的文字。以下範例說明如何偵測：

```
async for event in runner.run_async(
    user_id="test_user",
    session_id=session.id,
    new_message=content,
):
    if event.grounding_metadata:
        if event.grounding_metadata.grounding_chunks:
            for chunk in event.grounding_metadata.grounding_chunks:
                # Extract the encoded polyline from the maps chunk text
                if (
                    hasattr(chunk, "maps")
                    and chunk.maps
                    and hasattr(chunk.maps, "text")
                    and chunk.maps.text
                    and "Encoded Polyline" in chunk.maps.text
                ):
                    print("SUCCESS: Encoded Polyline detected in the response!")
```

步驟 9 · 約 2 分鐘

9. 清除所用資源

如要避免系統持續向您的 Google Cloud 帳戶收費，請刪除本程式碼研究室建立的資源。

1. 如果您為本程式碼研究室特地建立了專案，請刪除這整個專案：

```
gcloud projects delete $PROJECT_ID
```

如果您使用現有專案並想保留專案，則不需要刪除任何特定資源，因為 Agent 是在本機執行，而使用的 API 屬於無伺服器 API。

步驟 10 · 約 1 分鐘

10. 恭喜

恭喜！您已成功建立行程規劃代理程式，並使用 Google 地圖洞察資料做為基礎。

目前所學內容

- 如何使用 Agent Starter Pack 搭建新代理
- 如何將建立基準工具新增至 ADK 代理定義
- 如何使用內建的網路執行元件測試 ADK 代理

後續步驟

- 探索其他 ADK 工具和整合模式

參考文件

- [ADK 說明文件](#)
 - [Google 地圖基礎](#)
-